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Hybrid Deep Learning Framework for Adaptive Malicious Link Detection and Web Threat Intelligence

Combining Machine Learning, Deep Learning and Agentic AI for real-time URL threat classification

98.7% Detection accuracy	**4** URL classes: Benign, Phishing, Malware, Defacement	**7** Specialized AI agents + 1 orchestrator	**14** Lexical & security URL features

PROJECT TEAM

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ABSTRACT

Traditional URL detection relies on blacklists and fixed rules, which miss newly created malicious websites. This work presents a Hybrid Deep Learning Framework that combines a Random Forest model, a Deep Learning neural network and an Agentic AI architecture to classify URLs as Benign, Phishing, Malware or Defacement. Cooperating agents handle URL validation, feature extraction, threat detection, threat intelligence and report generation, and deliver a prediction, confidence score, risk level and security recommendations through a Flask web interface.

RESEARCH GAPS ADDRESSED

- Blacklists miss zero-day and newly created URLs
- Hand-crafted features limit ML detection power
- Existing tools classify only; no threat analysis
- Little automation of decisions and reporting
- Frequent retraining; limited adaptability

METHODOLOGY

User URL Input — Website URL submitted via Flask web interface

→ **Orchestrator Agent** — Coordinates every agent

→ I URL Validation · II Feature Extraction · III Random Forest · IV Deep Learning · V Hybrid Decision · VI Threat Intelligence

→ **Report Generation Agent** — Prediction, confidence score, risk level, threat category and recommendations shown on the web dashboard

LAYERED SYSTEM ARCHITECTURE

Presentation	HTML · CSS · JavaScript dashboard
Application	Flask web framework · URL input handling
Agent Layer	Orchestrator + 7 cooperating agents
ML / DL Models	Random Forest · Neural network (TensorFlow)
Data	Kaggle Malicious URLs dataset · 80:20 split

END-TO-END WORKFLOW

1. User submits URL
2. URL validation
3. Feature extraction
4. RF + DL prediction
5. Hybrid decision
6. Threat intelligence
7. Report generation
8. Result shown to user

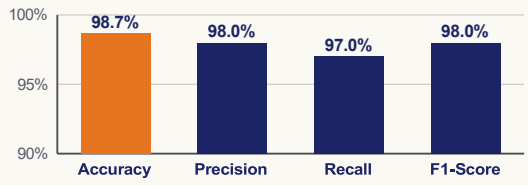
DATASET & PREPROCESSING

- Public Kaggle dataset: Benign, Phishing, Malware, Defacement URLs
- Duplicates and missing values removed
- Labels converted with label encoding
- Data shuffled to remove ordering bias
- 80:20 split for training and testing

URL FEATURES EXTRACTED

- URL length
- Number of dots
- Number of digits
- HTTPS usage
- Special characters
- Slash count
- URL entropy
- Domain length
- Number of hyphens
- Subdomains
- IP address present
- Suspicious keywords
- Query parameters
- Domain age

EVALUATION RESULTS



Bars scaled 90-100%. Intel Core i5, 8 GB RAM, Windows.

IMPLEMENTATION: WORKING PROTOTYPE



Home: overview of the AI Link Shield 2.0 system



Threat scanner: prediction, confidence and risk



Dashboard: scan statistics and model metrics

TECHNOLOGY STACK: Python 3 · Flask · Random Forest · Neural Network (TensorFlow) · scikit-learn · HTML · CSS · JavaScript · VS Code

CONCLUSION

Integrating Random Forest, Deep Learning and Agentic AI in one framework delivers accurate detection of Benign, Phishing, Malware and Defacement URLs (**98.7% accuracy**) while cutting false predictions. Automated validation, analysis and reporting give users clear, actionable guidance before they open a website.

LIMITATIONS & FUTURE WORK

Limits: accuracy depends on dataset quality and diversity; unseen threats need periodic retraining; deep models need more compute at scale.

Future work: Transformer and Graph Neural Network models, live threat-intelligence feeds, cloud security services, browser extension and mobile apps, and explainable AI (XAI).